**Example 7.7** From a solid cylinder whose height is 2.4 cm and diameter 1.4 cm, a conical cavity of the same height and base is hollowed out (Fig.7.13). Find the total surface area of the remaining solid.

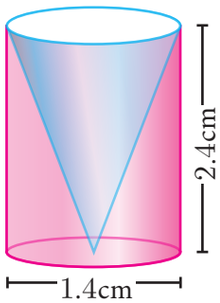


Fig. 7.13

***Solution*** Let $h$ and $r$ be the height and radius of the cone and cylinder.

Let $l$ be the slant height of the cone.

Given that, $h = 2.4$ cm and $d = 1.4$ cm ; $r = 0.7$ cm

$$\left.\begin{matrix}\text{Total surface area of the}\\ \text{remaining solid}\end{matrix}\right\} = \begin{matrix}\text{C.S.A. of the cylinder + C.S.A. of the cone}\\ \text{+ area of the bottom}\end{matrix}$$

$$= 2\pi rh + \pi rl + \pi r^2 \text{ sq. units}$$

$$= \pi r\left(2h + l + r\right) \text{ sq. units}$$

Now, $$l = \sqrt{r^2 + h^2} = \sqrt{0.49 + 5.76} = \sqrt{6.25} = 2.5 \text{ cm}$$

$$l = 2.5 \text{ cm}$$

$$\text{Area of the remaining solid} = \pi r\left(2h + l + r\right) \text{ sq. units}$$

$$= \frac{22}{7} \times 0.7 \times \left[\left(2 \times 2.4\right) + 2.5 + 0.7\right]$$

$$= 17.6$$

Therefore, total surface area of the remaining solid is 17.6 $cm^2$

**Progress Check**

1. Right circular cone is a solid obtained by revolving _____ about _____.
2. In a right circular cone the axis is _____ to the diameter.
3. The difference between the C.S.A. and T.S.A. of a cone is _____.
4. When a sector of a circle is transformed to form a cone, then match the conversions taking place between the sector and the cone.

| Sector | Cone |
|---|---|
| Radius | Circumference of the base |
| Area | Slant height |
| Arc length | Curved surface area |

### 7.2.4 The Sphere

**Definition :** A sphere is a solid generated by the revolution of a semicircle about its diameter as axis.



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